pyDMS

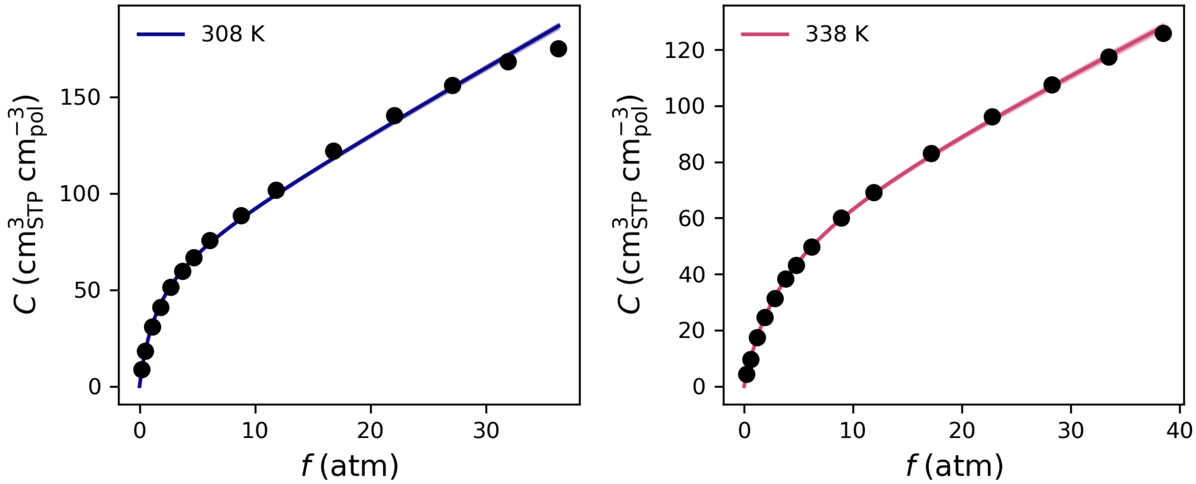
Reproducible Dual-Mode Sorption Parameters

Analysis performed at 2025-06-13 11:21:49 Eastern Daylight Time

DMS Parameters

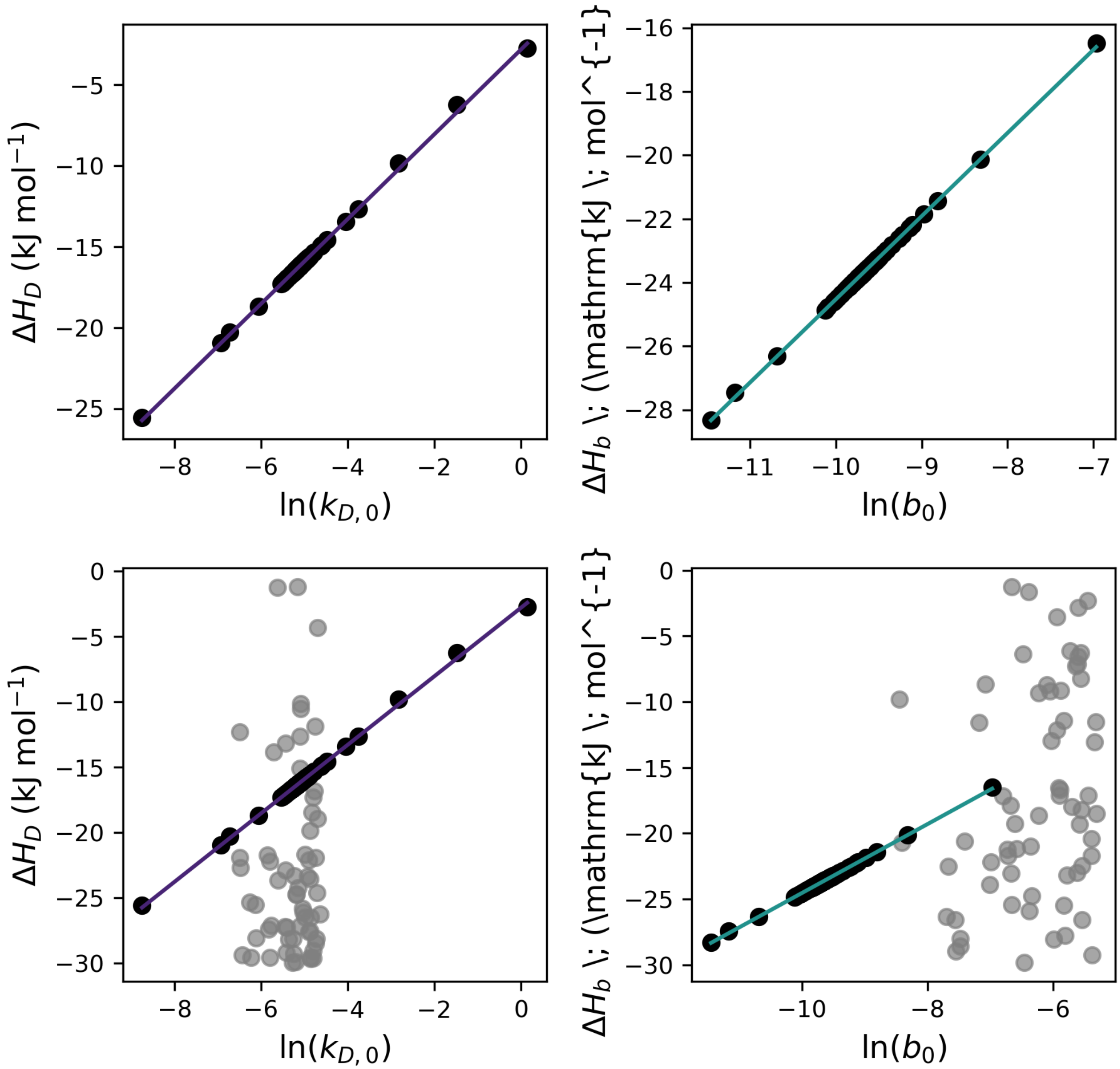
Temperature	Parameter	Result ± Error
308 K	k_D	3.370346214467855 ± 0.07310618303203206
	C'_H	67.18687837313098 ± 0.495083055473728
	b	0.652576273740172 ± 0.013566895161178929
338 K	k_D	1.9211623811308824 ± 0.061267001342491155
	C'_H	59.07542246174292 ± 0.6961680594956547
	b	0.2876570024646488 ± 0.006297662286261411

Sorption Isotherms

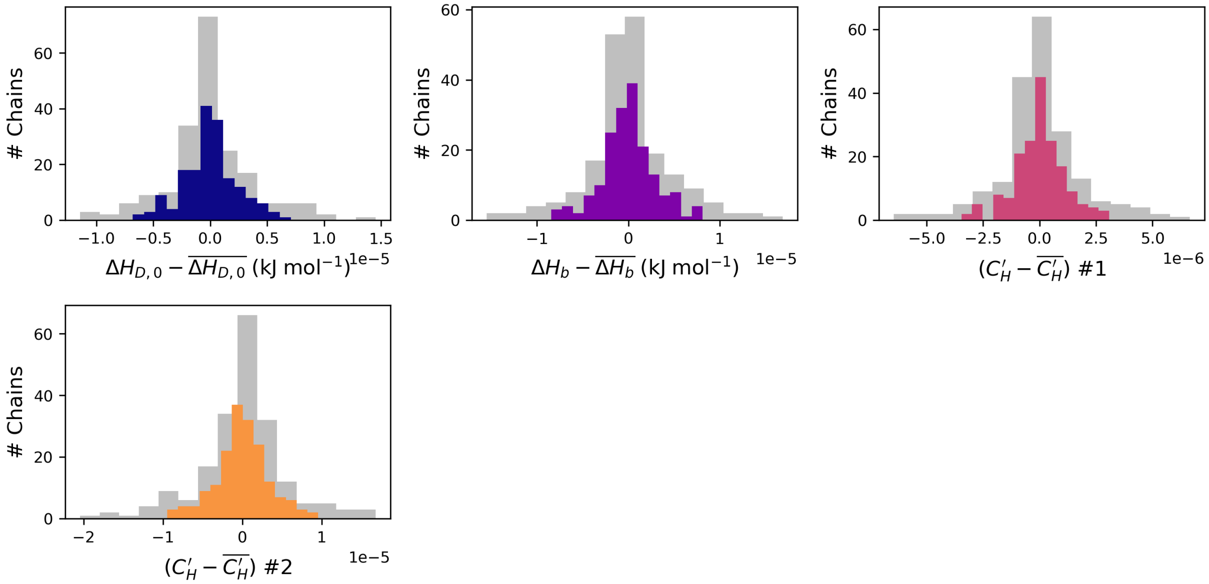


Linear Free Energy Relationships

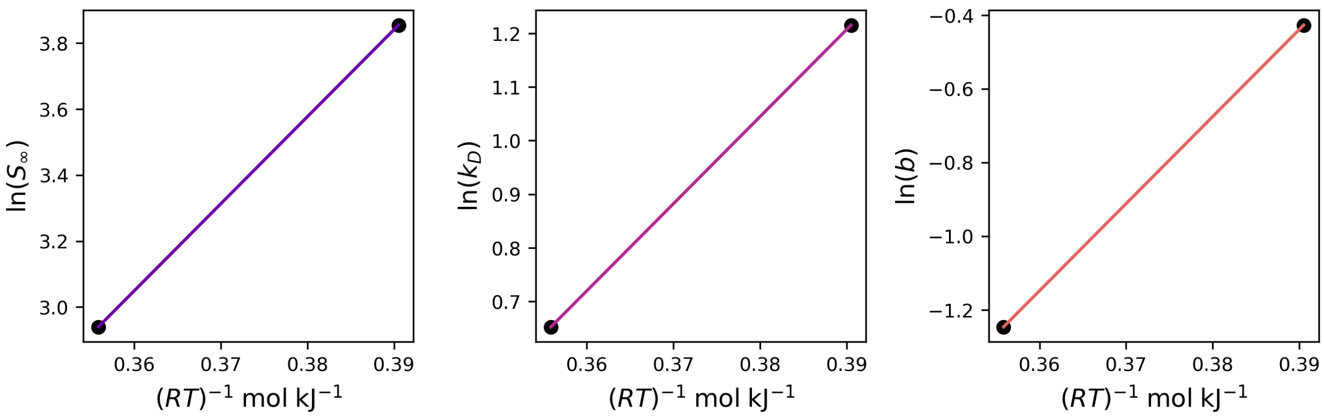
Heat of Henry Sorption	$\Delta H_D = -2.819 + 2.6178 \ln(k_{D,0})$
Heat of Langmuir Sorption	$\Delta H_b = 1.6306 + 2.6164 \ln(b_0)$



van't Hoff Relationships



Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	1.578e-03±inf	-26.391873 ± inf
i = Langmuir	5.989e-03 ± inf	-16.216541 ± 0.946022
i = Henry	6.404e-05 ± inf	-23.633308 ± 0.274494

Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-1, -30]
ΔH_b Initial Guess Range	[-1, -30]
$k_{D,0}$ Initial Guess Range	[0.001, 0.01]
b_0 Initial Guess Range	[0.0001, 0.005]
ΔH_D Solver Bounds	[-50, 0]
ΔH_b Solver Bounds	[-50, 0]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H #1	[0 150]
C'_H #2	[0 150]
Trials	200
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-12
xtol (trust-constr)	1e-12
gtol (trust-constr)	1e-12

Questions/Comments/Concerns?

Documentation available through ReadTheDocs (*)

Please reach out to Brandon C. Tapia via email (bctapia@mit.edu) or through the pyDMS GitHub page

Code Availability

pyDMS can be downloaded via:

- GitHub (source code): git clone **(insert link)
- pip: pip install **insert name
- Anaconda: conda install **insert name

Citation

If you used pyDMS in any presented or published work please cite:

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